

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
8	(a)	(i)		Skin (connective tissue) / skin flora (bacteria)/ blood clotting / inflammation/ phagocytes / macrophage lysozyme/ stomach acid/ low pH in stomach Any 3 for 1 mark	1			1		
		(ii)		{Many organisms/ named organism} live {in/on} the (human) body	1			1		
		(iii)		Cell lysis / cell bursts	1			1		
		(iv)		Any three × (1) from: • {Macrophages/ phagocytes} {engulf virus infected cell / carry out phagocytosis}/ antigen presentation (1) • clonal expansion/ T cells {proliferate/divide many times} / (1) • Cytokines stimulate phagocytosis (1) • Killer T cells cause cell lysis (1)	1	2		3		
	(b)	(i)		• Antibiotics can act on {cell wall of bacteria/ or description of/ hydrolyse peptidoglycan} (1) • Antibiotics affect {metabolism/protein synthesis} by bacterial cell (1) • Viruses don't have a {cell wall / metabolism} so not affected by antibiotics (1)	2		1	3		
		(ii)		7250 (mg) (2) If incorrect award 1 mark for any combination of three parts of equation 25mg × 14.5 × 2 × 10		2		2	2	

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		(iii)		Amoxycillin is bactericidal + {as the bacterial numbers are reduced / has killed bacteria} (1) Tetracycline is bacteriostatic + {as it has prevented replication/no cell death/cell numbers remain constant }(1)			2	2		
		(iv)		Any two × (1) from: Same {concentration/ dosage} of antibiotics (1) same composition of {culture media/agar}/ named constituent of {culture media/agar} (1) Same strain of <i>S. pyogenes</i> (1) pH (1) NOT incubation temperature/ oxygen		2		2		2
		(v)		Human pathogen / (close to human) body temperature		1		1		1
	(c)	(i)		- May be {rapid/ high} mutation rate in rhinovirus /antigens change/viruses are antigenically labile (1) - Many {strains/serotypes/ antigenic types} of rhinovirus (1) - Vaccines would need to contain antigens from all strains (to provide complete immunity) (1) - Each antigen must produce a strong immune response/be highly antigenic / be immunogenic(1)		1	2	3		
		(ii)		Religious objections/ safety fears/ side effects/ individual rights/ some people may be too ill for vaccination e.g. due to HIV-AIDS or Chemotherapy		1		1		
				Question 8 total	6	9	5	20	2	3